

Evidence-first selective inference for Bengali hallucination detection

A study of routing, abstention, provenance, and domain shift

Research question

When should exact evidence decide, and when should the system abstain to a language-model judge?

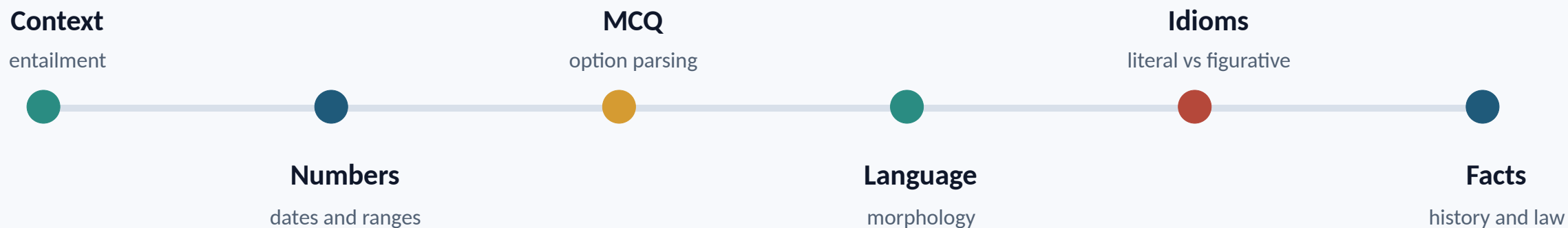
We trace the hypotheses, failed experiments, routing policies, regression gates, and Phase 2 shift.

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One binary label hid many different reasoning tasks

The same label had to cover context faithfulness, language knowledge, and outside facts.

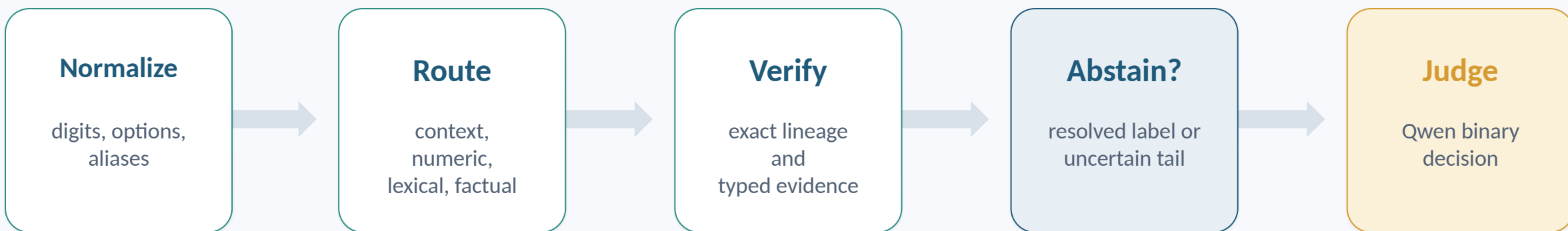


A single judge prompt treated all six as the same problem.

That worked as a baseline, but it was prompt-sensitive and weak on culturally specific relations.

The working hypothesis became a selective cascade

Every stage could answer, or abstain. Abstention was a normal outcome, not a failure.



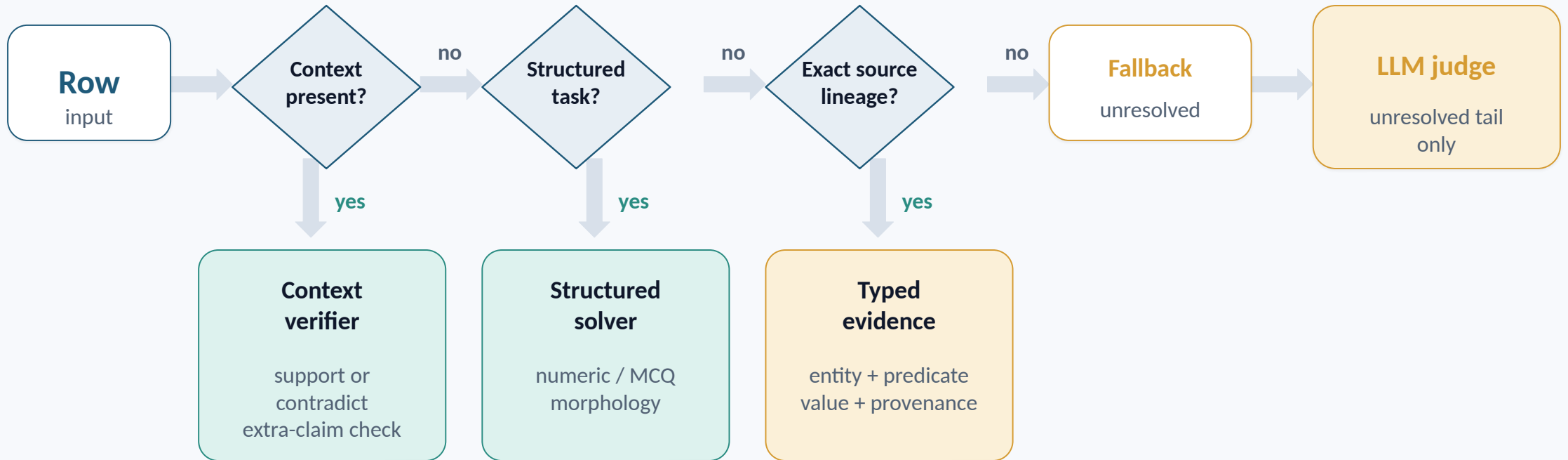
Technical contributions

Selective routing • task solvers • claim verification • typed provenance • offline T4×2

Every deterministic decision records its source family and requested relation.

Routing made abstention an explicit state

A branch could return 0, 1, or abstain. Only abstentions moved to the next branch.

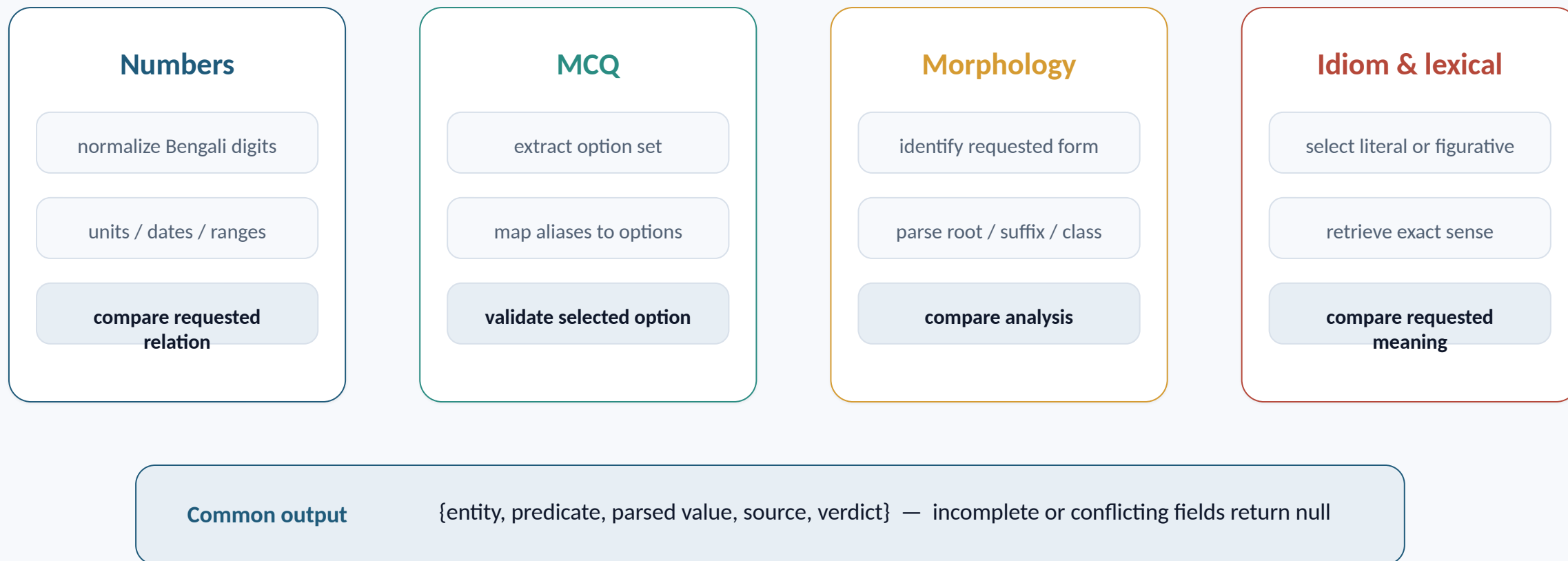


Decision contract

`branch(row) -> {0, 1, null}; null means abstain, never a guessed label`

Structured solvers shared one decision contract

Each solver parsed a task-specific representation, verified the requested relation, then answered or abstained.



A correct fact can still answer the wrong relation

Released sample 4 contains both the film title and year; the prompt asks specifically for the year.

Released labeled example

Context (English)
Runway, Tareque Masud's latest feature film, was released in 2010.

Prompt
In which year was the film released?

Response Runway

Released label: 0

→

Relation check

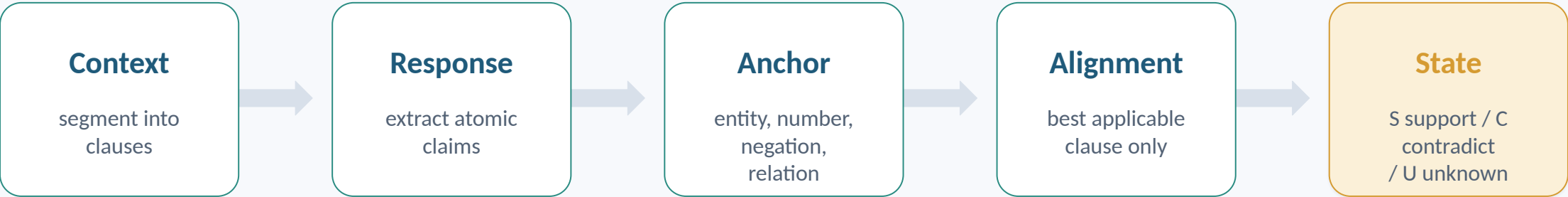
Entity	Runway	matches
Requested predicate	release_year	expected
Response type	film_title	observed

predicate mismatch → label 0

Similarity finds a related answer. The relation gate rejects it.

Context rows were reduced to claim-level entailment tests

The verifier compared atomic response claims against the relevant context clause, not against global word overlap.



Decision rule over response claims $c_1 \dots c_n$

Any contradiction	exists i : $\text{state}(c_i) = C$	label 0
Complete support	for all i : $\text{state}(c_i) = S$; no extra factual claim	label 1
Incomplete alignment	otherwise	abstain

Qwen judged 158 rows, not the whole test set

The fallback was short, deterministic, and biased toward catching hallucinations.

Context row

1 entailment view

Compare the response against the supplied passage.

Any contradiction returns 0.

No-context row

Up to 3 short views

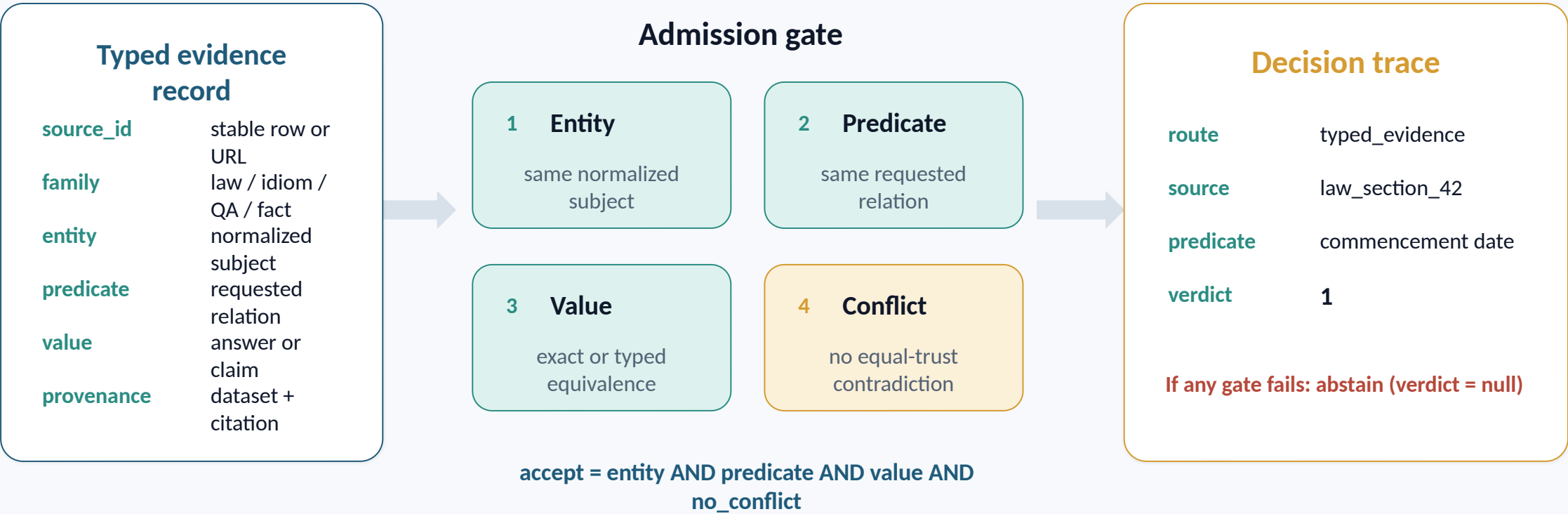
Fact check | exam grading | zero-bias check

Label 1 requires unanimous positive votes.

Qwen3.6-27B Q4_K_P | one-token grammar | fixed seeds and batches

A source record was accepted only after four fields aligned

The trace records why a row was resolved, its source, and the relation that was tested.



Versioned comparisons locate the main design pivots

These are cumulative versions, not controlled single-factor ablations.

Version	Public F1	Rows judged	Main methodological change
Baseline	0.657	none	character and embedding reference
Uniform Qwen	0.797	all rows	general reasoning improved; one policy remained
v6	0.903	reduced	source and structured routing
v37	0.961	312	stricter provenance gate
v85	0.966	193	reproducible two-source anchor
v91	0.967	158	typed idiom and predicate guards

The largest observed step coincided with routing: 0.797 → 0.903.

Decision logs turned failed runs into routing rules

Each negative result changed what later versions were allowed to decide.

Experiment	Observed result	Decision recorded
Prompt voting	0.851 local, 0.783 public	Stop prompt churn; require new evidence
TF-IDF verifier	0.826 out of fold	Keep as a portable fallback, not a scorer
Raw QA retrieval	Relevant passage, wrong relation	Retrieval proposes; it never decides
v87 fact edits	0.966 to 0.965 public	Require source + predicate; revert

Admission rule: a change must add an evidence trace and preserve the regression suite.

[7] Rejected and scored experiment records in EXPERIMENT.log

Reproducibility was treated as an evaluation result

Correctness, deterministic execution, output integrity, and runtime were checked separately.

249

automated regression tests

3

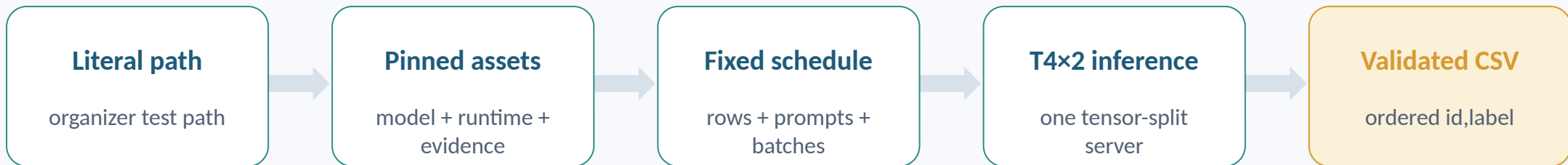
byte-identical inference runs

225

deterministic controls

0

schema or ID-order failures



3 runs matched byte-for-byte • scored CSV hash matched • 5,000 Phase 2 rows validated in 3:52:02

Phase 2 completed in 3 h 52 min with 95.9% judged

Measured organizer-run inference on 5,000 rows using Kaggle Tesla T4 x2.

5,000

held-out rows

207

deterministic (4.14%)

4,793

Qwen judge (95.86%)

3:52:02

validated end to end

Measured milestones	Elapsed	Observation
Router complete	1:13.18	207 resolved; 4,793 abstained
Model ready	2:51.25	16.332 GiB Qwen across two T4 GPUs
Judge complete	3:52:01.91	13,747.50 s; 0.349 rows/s

42.97% of the 9-hour budget used

5:07:58 headroom

Phase 2 source shift collapsed deterministic coverage

Exact Phase 1 lineage will cover less. The fallback must remain safe when coverage drops.

Web and news

Common Crawl, recent newspapers

Knowledge

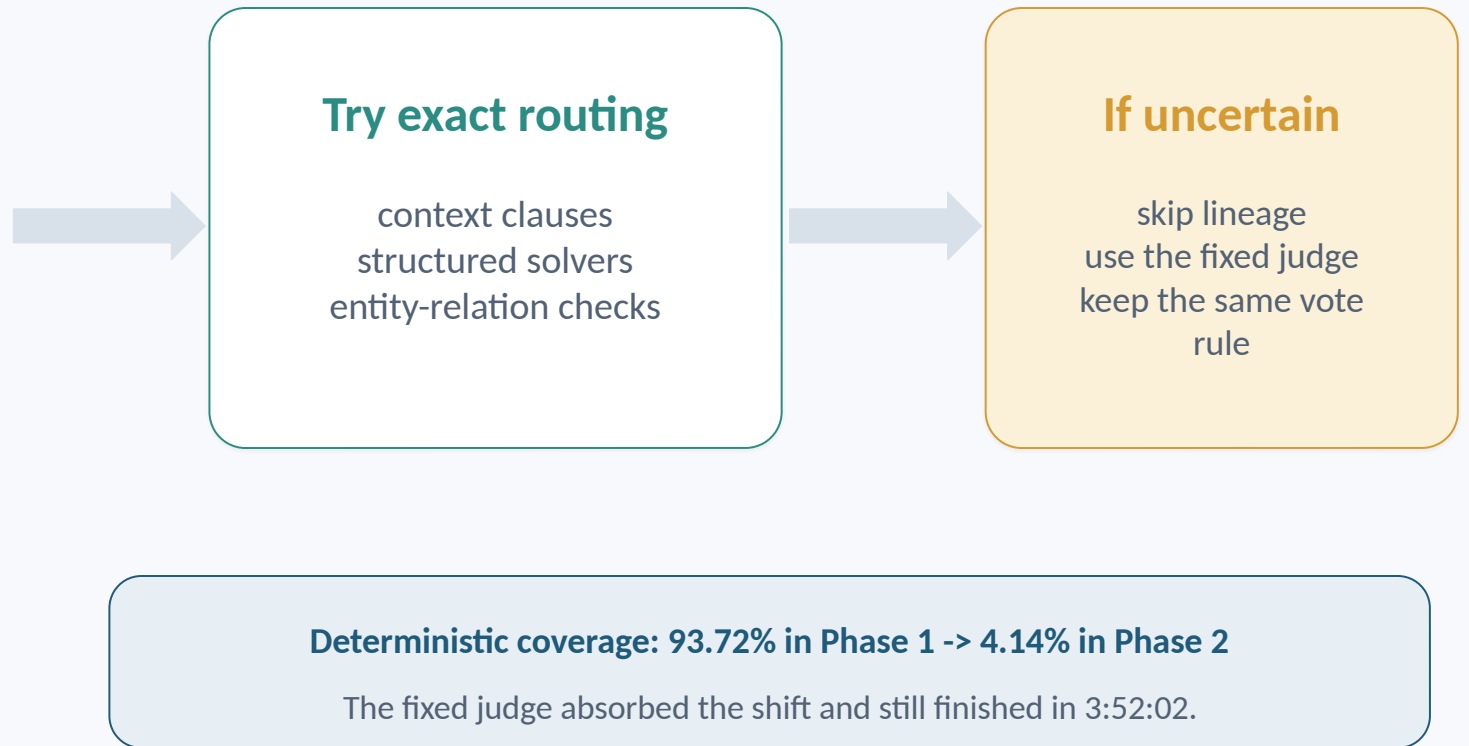
Wikipedia and Banglapedia

Civic and legal

government services and bdlaws

Education and culture

NCTB textbooks and literature



What the experiments support— and what remains open

Supported

Task families need separate policies.
Predicate alignment prevents relation
swaps.
Abstention protected runtime under
shift.

Limitations

Only 299 released labels.
Aggregate splits provide no row-level
errors.
Phase 2 accuracy is still unknown.

Next analyses

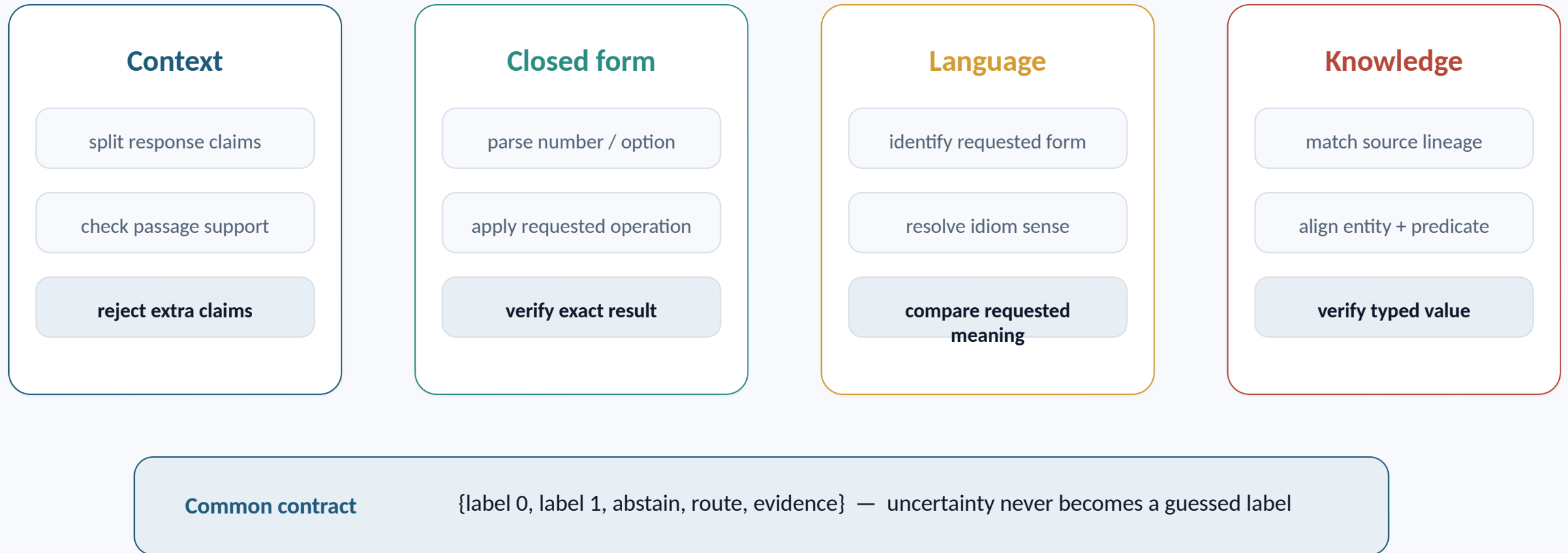
Inspect Phase 2 errors when released.
Calibrate the judged tail.
Expand licensed civic and news
evidence.

Questions

Selective inference remained reproducible and completed the held-out run within the runtime limit.

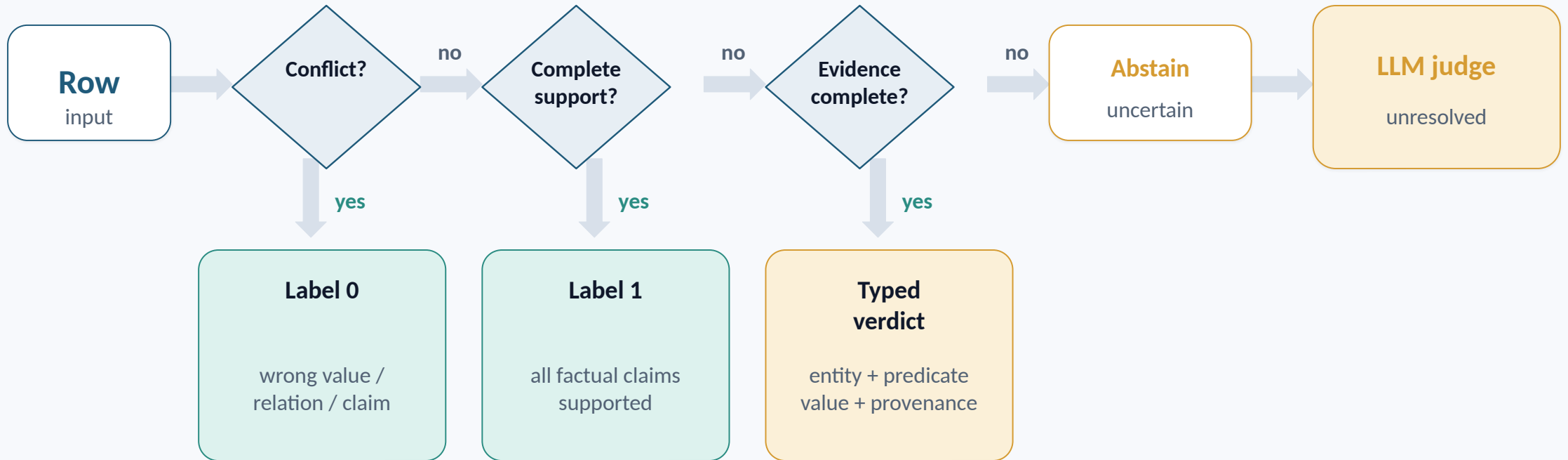
Evidence standards changed with the row type

The binary label stayed fixed; the support test changed with the task and available evidence.



Decision precedence kept uncertainty out of labels

Contradiction can decide zero; complete support can decide one; every ambiguous case abstains.

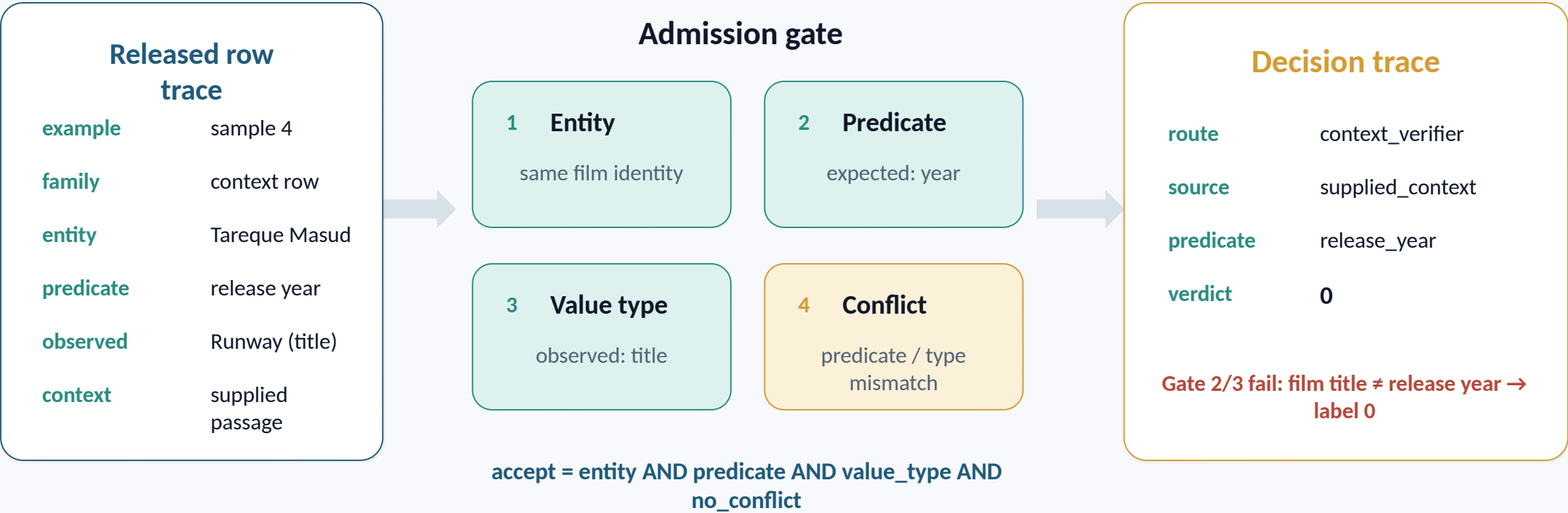


Precedence contract

contradiction → 0; complete support → 1; exact typed evidence may decide; otherwise null → judge

One trace exposes the exact reason behind a verdict

Released example: the response names the film when the prompt asks for its release year.



Five evidence layers answer five different questions

Only labeled rows measure classification quality; software and runtime checks establish other properties.

299

released labeled rows

249

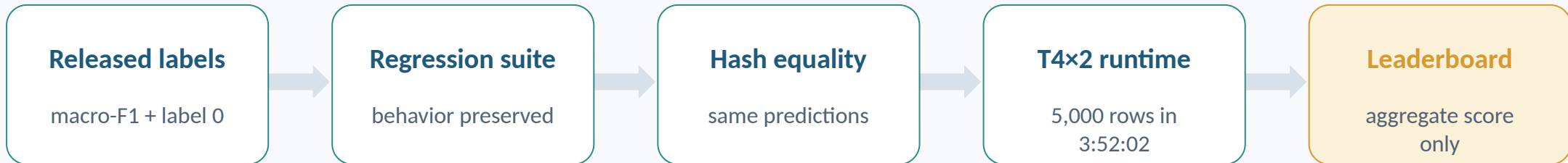
automated regression tests

3

byte-identical inference runs

0

schema or ID-order failures



Accuracy ≠ reproducibility ≠ output integrity ≠ runtime; each claim needs its own evidence.

Phase 2 defines a focused next-experiment queue

Observed shift

Deterministic coverage: 93.72% → 4.14%.
Judge handled 4,793 / 5,000 rows.
Accuracy is not yet released.

Next experiments

Label and cluster judged-tail errors.
Calibrate abstention and judge votes.
Add licensed civic and news evidence.

Rejected shortcuts

Do not force fuzzy matches.
Do not fine-tune on only 299 labels.
Do not call runtime an accuracy result.

Decision

Keep the conservative admission contract; optimize the judge tail only after Phase 2 error labels exist.